Atanu Saha

MSc in Big Data Analytics | BSc(Hons) in Statistics
Location: Pune, India

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Education

- **MSc in Big Data Analytics** — *Aug 2023 – Jul 2025*
  Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah — CGPA: 7.07
- **BSc(Hons) in Statistics** — *Aug 2020 – Jul 2023*
  Kalyani Mahavidyalaya, University of Kalyani — CGPA: 8.5/10
- **Higher Secondary(Science)** — *2018 – 2020*
  Kanchrapara Harnett High School — Percentage 93.6%
- **Secondary(10th)** — *2006 - 2018*
  Bedibhawan Rabitirtha Vidyalaya — Percentage 90.14%

Experience

- **Research Fellow** — *May 2026 – Present*
  CVIT, IIIT Hyderabad, India
  - Working under the supervision of **Prof. C.V. Jawahar.**
- **Visiting Faculty** — *Aug 2025 – April 2026*
  FLAME University, Pune, India
  - Taught Precalculus and Calculus courses and currently teaching Time Series Analysis and Forecasting, focusing on both theoretical foundations and practical applications.
- **Research Assistant** — *June 2025 – Dec 2025*
  FLAME University, Pune, India
  - Created a novel **Drum Sheet Music dataset** for document image analysis and Optical Music Recognition (OMR).
  - Worked on analysis of drum notation for document understanding and **communicated a research paper to ICDAR 2026**.
  - Conducted research on music notation analysis under the supervision of Dr. Chiranjoy Chattopadhyay.
- **Intern** — *January 2025 – April 2025*
  FLAME University, Pune, India
  - Contributed to the development of a **VR-based piano learning environment** integrating Optical Music Recognition (OMR).
  - Co-authored the paper *"From Notes to Keys: A VR Learning Environment for Sheet Music Interpretation"*, **accepted at ICDAR 2025**.
- **Subject Matter Expert – Statistics & Probability** — *Dec 2022 – Nov 2024*
  Chegg India | Remote
  - Solved and reviewed advanced-level questions in Statistics, Probability Theory, and Data Analysis.
  - Provided step-by-step solutions aligned with university standards while ensuring academic integrity.
  - Maintained high CF Score and high accuracy.

Achievements

- **CR Rao Statistics Olympiad:** Cleared with Rank 4 (AIR 2) — *2020*
- **ISI MSQMS Entrance:** Cracked written exam and interview (AIR 25) — *2023*
- **SVMCM Scholarship:** Received scholarship from Govt. of West Bengal — *2018-2025*

Publication

- **From Notes to Keys: A VR Learning Environment for Sheet Music Interpretation** — *2025*
  Accepted at ***ICDAR 2025 (International Conference on Document Analysis and Recognition)***
  - Proposed a VR-based piano learning system integrating Optical Music Recognition (OMR) and immersive visual feedback using Meta Quest 3.
  - Achieved high accuracy in music note detection with scale and rotation invariance, improving sheet music interpretation for beginners.

Projects

- **Curvature-Aware Data Augmentation for Steel Surface Defect Detection via Physics-Informed Image Re-projection** *2025*
  - Proposed a physics-informed data augmentation pipeline using calibrated camera re-projection and physically based rendering to generate realistic non-flat defect images.
  - Developed a Python-based parametric framework controlling curved geometries, steel materials, and illumination to ensure geometric and photometric plausibility. Showed improved defect detection performance on curved steel surfaces while preserving accuracy on planar benchmarks.
  - Applied Grad-CAM to visualize and interpret model predictions, highlighting salient regions responsible for defect detection under varying geometric conditions.

- **Classification of Ragas in Indian Classical Music Using Machine Learning Algorithms** *2024*
  - Created a structured dataset comprising audio recordings of 9 different Indian classical ragas for supervised learning.
  - Performed audio preprocessing and feature extraction using the *librosa* library, including MFCCs, chroma features etc.
  - Implemented and evaluated multiple machine learning models including KNN, SVM, Random Forest, and Naive Bayes for raga classification.
  - Achieved **95% classification accuracy** using the Random Forest classifier, demonstrating strong performance for automated raga identification.

Social Work

- **Free Tuition Project (COVID-19 Support Program)** *2021 – Present*
  Founder & Volunteer Teacher
  - Initiated and conducted a free tuition program during the COVID-19 pandemic starting in 2021 to support underprivileged students affected by school closures.
  - Provided academic support to students from **Class 8 to Class 12** across all streams — Science, Arts, and Commerce
  - Mentored and guided students academically and psychologically during the pandemic, ensuring continuity of education in challenging circumstances.
  - One of the mentored students became a **rank holder in the West Bengal State Board examinations**.

Technical Skills and Interests

Areas of Interest: Computer Vision, Machine Learning, Deep Learning, Artificial Intelligence
Languages: Python, C, R

Strengths

Hardworking | Eye for detail | Creative | Innovative Thinker | Uplifts Others